Stochastic Analysis

LectureNotes

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Chapter 1

Intro

Riemann-Stieltjes Integral: If f is continuous and g is of finite variation then $\int_a^b f(x)\ \mathrm{d}g(x)$ is well-defined and is defined to be

$$\int_a^b f(x)\ \mathrm{d}g(x) := \lim_{n\to\infty} \sum_{i=0}^{n-1} f(\xi_i)\left(g(x_{i+1}) - g(x_i)\right)$$

with $a \le x_0 < x_1 < \cdots < x_n \le b$ and $\xi_i \in [x_i, x_{i+1}]$.

Main Task: Itô integral $(H \cdot X)_t = \int_0^t H_s\ \mathrm{d}X_s$.

E.g X is Brownian motion.

Theorem 1.0.1. $\displaystyle\sum_{k=1}^{2^n} \left(B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}}\right)^2 \xrightarrow{a.s.} 1.$

Finite Variation of a function $g : [a, b] \to \mathbb{R}$ is

$$V_g([a, b]) = \int_a^b |\mathrm{d}g(x)| = \sup_n \sum_{i=1}^{n} |g(x_{i+1}) - g(x_i)|$$

Corollary 1.0.1. $V_B([0, 1]) = \infty$ a.s.

Proof for Corollary.

■.

$$\begin{aligned}(\star) \sum_{k=1}^{2^n} \left(B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}}\right)^2 &\le \sup_{1\le k\le 2^n} \left|B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}}\right| \cdot \sum_{k=1}^{2^n} \left|B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}}\right| \\ &\le V_B([0, 1]) \cdot \sup_{1\le k\le 2^n} \left|B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}}\right|\end{aligned}$$

If $V_B([0, 1]) < \infty$, then $(\star) \to 0$.

Contradiction with $(\star) \to 1$, so w.p.1 $V_B([0, 1]) = \infty$. □

$(H \cdot X)_t = \int_0^t H_s\ \mathrm{d}X_s$, where H_s is previsible or predictable, X_s is semi martingale.

Example . Let $H_n \in \mathcal{F}_{n-1}$, (M_n) be a mart.

Then $(H \cdot M)_n = \sum_{k=1}^n H_k(M_k - M_{k-1})$ is a martingale.

1.1 Set-up

Given a probability space $(\Omega, \mathcal{F}, \mathsf{P})$. A stochastic process $X = (X_t)_{t\geq 0}$ is a collection of random variables taking values in $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$.

$$\begin{aligned} X\colon [0,\infty) \times \Omega &\to \mathbb{R}^d \\ (t,\omega) &\mapsto X_t(\omega). \end{aligned}$$

Fix ω, then the function $t \mapsto X_t(\omega)$ is called a sample path.

Definition 1.1.1. For two stochastic process X and Y, we say Y is a modification of X if

$$\forall\ t \geq 0, \quad \mathsf{P}(X_t = Y_t) = 1$$

Definition 1.1.2. We say X and Y are indistinguishable if

$$\mathsf{P}(X_t = Y_t,\ \forall\ t \geq 0) = 1.$$

And then Y is a version of X.

Example . Let Y be a modification of X and suppose X and Y both have a.s. right continuous sample paths. Then X & Y are indistinguishable.

Definition 1.1.3. A function f is right-continuous if $f(x) = f(x+) = \lim_{t\downarrow x} f(t)$.

Proof. Since $\forall\ t \geq 0$, $\mathsf{P}(X_t \neq Y_t) = 0$, we get

$$\mathsf{P}\left(\bigcup_{q\in\mathbb{Q}^+} X_q \neq Y_q\right) \leq \sum_{q\in\mathbb{Q}^+} \mathsf{P}(X_q \neq Y_q) = 0.$$

$$\Rightarrow\ \mathsf{P}\left(\bigcap_{q\in\mathbb{Q}^+} X_q = Y_q\right) = 1.$$

For two right-continuous functions f and g, if $f(q) = g(q)$, $\forall\ q \in \mathbb{Q}^+$, then $f(t) = g(t)$, $\forall\ t \geq 0$.

Hence $\mathsf{P}\left(X_t = Y_t,\ \forall\ t \geq 0\right) = 1$. □

Example . Let $T \sim \mathrm{Exp}(1)$. Define

$$\begin{aligned} X_t &\equiv 0, \quad \forall\ t \geq 0 \\ Y_t &= \begin{cases} 0, & t \neq T \\ 1, & t = T. \end{cases} \end{aligned}$$

$\forall\ t \geq 0$, $\mathsf{P}(X_t \neq Y_t) = \mathsf{P}(Y_t = 1) = \mathsf{P}(t = T) = 0$.

1.2 Filtration

Let $(\mathcal{F}_t)_{t\geq 0}$ be a filtration of $(\Omega, \mathcal{F})$ such that $\forall\ 0 \leq s < t < \infty$, $\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}$.

Example . (Natural Filtration) Let $(X_t)_{t\geq 0}$ be a stochastic process, Define

$$\mathcal{F}_t^X = \sigma(X_s, 0 \leq s \leq t).$$

Then $(\mathcal{F}_t^X)$ is a filtration.

Obviously, $X_t \in \mathcal{F}_t^X$.

Exercise 1.7. Let X be a process, every sample path of which is <u>RCLL</u>. Let

$$A = \{\omega \colon X(\omega) \text{ is continuous on } [0, t_0)\}$$

for some fixed $t_0 > 0$. Show that $A \in \mathcal{F}_{t_0}^X$.

<u>RCLL</u>: $f(x) = f(x+)$ and $f(x-)$ exists.

Proof.

$$\begin{aligned} A &= \{\omega \colon \forall\ s \in (0, t_0),\ X(s-, \omega) = X(s, \omega)\} \\ &= \{\omega \colon \forall\ q \in (0, t_0) \cap \mathbb{Q},\ X(q-, \omega) = X(q, \omega)\} \end{aligned}$$

Let $(0, t_0) \cap \mathbb{Q} = (q_m)_{m\geq 1}$, then

$$\begin{aligned} A &= \left\{\omega \colon \forall\ q_m \text{ with } m \geq 1,\ \forall\ k \geq 1,\ \exists\ N \geq 1,\ \forall\ n \geq N, \left|X(q_m - \frac{1}{n}, \omega) - X(q_m, \omega)\right| < \frac{1}{k}\right\} \\ &= \bigcap_{m=1}^{\infty} \bigcap_{k=1}^{\infty} \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} \left\{\left|X(q_m - \frac{1}{n}, X(q_m)) < \frac{1}{k}\right|\right\} \in \mathcal{F}_{t_0}^X. \end{aligned}$$

□

Exercise 1.8. Let X be a process whose sample paths are RCLL a.s. Then $A \notin \mathcal{F}_{t_0}^X$ is possible.

Proof. [TODO] □

Definition 1.2.1. A filtration $\{\mathcal{F}_t\}_{t\geq 0}$ is said to satisfy the <u>usual condition</u> if

(a) $\mathcal{F}_t = \mathcal{F}_{t+} = \bigcap_{s>t} \mathcal{F}_s$;

(b) $\mathcal{F}_0$ contains all the <u>$\mathsf{P}-$negligible</u> sets in $\mathcal{F}$.

Definition 1.2.2. An even A is P-negligible if $\exists\ N \in \mathcal{F}$ s.t. $A \subseteq N$, $\mathsf{P}(N) = 0$.

Definition 1.2.3. $\mathcal{F}_{t+} = \bigcap_{s>t} \mathcal{F}_s$, $\mathcal{F}_{t-} = \sigma\left(\bigcup_{s<t} \mathcal{F}_s\right)$. We say a filtration is right-continuous if $\mathcal{F}_t = \mathcal{F}_{t+}$.

Definition 1.2.4. The stochastic process X is adapted to the filtration $\{\mathcal{F}_t\}_{t\geq 0}$ if $\forall\ t \geq 0$, $X_t \in \mathcal{F}_t$.

Obviously (X_t) is adapted to the natural filtration.

Definition 1.2.5. The stochastic process X is called progressively measurable with respect to $(\mathcal{F}_t)$ if $\forall\ t \geq 0$, the map

$$(s,\omega) \mapsto X_s(\omega)$$
$$([0,t] \times \Omega, \mathcal{B}([0,t]) \otimes \mathcal{F}_t) \to (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$$

is measurable.

adapted $\supseteq$ progressively measurable $\supseteq$ previsible

Proposition 1.2.1. If X is adapted to the filtration $(\mathcal{F}_t)$ and every sample path is right-continuous or left-continuous, then X is also progressively measurable.

Proof for Proposition.

■. Assume right-continuity $\forall\ t > 0$, $\forall\ n \geq 1$, $\forall\ 0 \leq k \leq 2^n - 1$,

$$X_s^{(n)}(\omega) = X_{\frac{k+1}{2^n}t}(\omega), \quad \forall\ \frac{kt}{2^n} < s \leq \frac{k+1}{2^n}t$$

Then $\lim\limits_{n\to\infty} X_s^n(\omega) = X_s(\omega)$ for all $(s,\omega) \in [0,t] \times \Omega$.

It suffices to prove

$$(s,\omega) \mapsto X_s^{(n)}(\omega)$$
$$([0,t] \times \Omega, \mathcal{B}([0,t]) \otimes \mathcal{F}_t) \to (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$$

is measurable.

$\forall\ A \in \mathcal{B}(\mathbb{R}^d)$,

$$\begin{aligned}&\left\{(s,\omega)\colon X_s^{(n)}(\omega) \in A\right\}\\ =&\bigcup_{k=0}^{2^n-1}\left(\frac{k}{2^n}t, \frac{k+1}{2^n}t\right] \times \left\{X_{\frac{k+1}{2^n}t} \in A\right\} \in \mathcal{B}([0,t]) \otimes \mathcal{F}_t\end{aligned}$$

We get $X_s^{(n)}(\omega)$ is measurable. Hence $X_s = \lim_{n\to\infty} X_s^{(n)}$ is also measurable. □

1.3 Stopping Times

$(\Omega, \mathcal{F}, \mathsf{P})$ and $(\mathcal{F}_t)_{t\geq 0}$.

Definition 1.3.1. A random time T is a stopping time of $(\mathcal{F}_t)$ if $\forall\ t \geq 0$, $\{T \leq t\} \in \mathcal{F}_t$.

Definition 1.3.2. A random time T is an optional time of $(\mathcal{F}_t)$ if $\forall\ t \geq 0$, $\{T < t\} \in \mathcal{F}_t$.

Proposition 1.3.1. stopping time $\subseteq$ optional time and stopping $=$ optional if the filtration is right-continuous.

Proof for Proposition.

- "$\Rightarrow$" $\{T < t\} = \bigcup_{n=1}^{\infty}\{T \le t - \frac{1}{n}\} \in \mathcal{F}_t$.
  "$\Leftarrow$" If T is an optional time, then $\forall\ m \ge 1$,

$$\{T \le t\} = \bigcap_{n=m}^{\infty}\left\{T < t + \frac{1}{n}\right\}.$$

Because $\{T < t + \frac{1}{n}\} \in \mathcal{F}_{t+\frac{1}{n}} \subseteq \mathcal{F}_{t+\frac{1}{m}}$.

$\Rightarrow\ \{T \le t\} \in \mathcal{F}_{t+\frac{1}{m}}$

$\Rightarrow\ \{T \le t\} \in \bigcap_{m=1}^{\infty}\mathcal{F}_{t+\frac{1}{m}} = \mathcal{F}_{t^+} = \mathcal{F}_t$. □

Lemma 1.3.1. If T and S are stopping times, then $T \wedge S$, $T \vee S$, $T + S$ are stopping times.

Proof. $\forall\ t \ge 0$,

1. $\{T \wedge S \le t\} = \{T \le t\} \cup \{S \le t\} \in \mathcal{F}_t$.
2. $\{T \vee S \le t\} = \{T \le t\} \cap \{S \le t\} \in \mathcal{F}_t$.
3. $\{T + S \le t\} \in \mathcal{F}_t \Leftrightarrow \{T + S > t\} \in \mathcal{F}_t$.

$$\begin{aligned}
&\{T + S > t\}\\
=&\{T = 0,\ S > t\} \cup \{S = 0,\ T > t\} \cup \{0 < T < t,\ T + S > t\} \cup \{S > 0,\ T \ge t\}\\
\in&\mathcal{F}_t \cup \mathcal{F}_t \cup \{0 < T < t,\ T + S > t\} \cup \mathcal{F}_t.
\end{aligned}$$

It suffices to show

$$\{0 < T < t,\ T + S > t\} = \bigcup_{\substack{r \in \mathbb{Q}^+\\ 0<r<t}} \{r < T < t,\ S > t - r\}.$$

Because the right term belong $\mathcal{F}_t$.

"$\Leftarrow$ " $S > t - r > t - T$, so $S + T > t$.

"$\Rightarrow$ " $\exists\ r \in (0, T)$, $S + r > t$ since $S + T > t$.

□

Lemma 1.3.2. If $\{T_n\}_{n\ge1}$ are stopping times, then $\sup_{n\ge1} T_n$ is a stopping time.

Proof. $\forall\ t \ge 0$, $\{\sup_{n\ge1} T_n \le t\} = \bigcap_{n=1}^{\infty}\{T_n \le t\} \in \mathcal{F}_t$. □

Definition 1.3.3. The stopping σ-algebra is $\mathcal{F}_T = \{A \in \mathcal{F} \mid \forall\ t \ge 0, A \cap \{T \le t\} \in \mathcal{F}_t\}$.

Proposition 1.3.2. If $T(\omega) = t_0$ for some fixed $t_0 \ge 0$, $\forall\ \omega \in \Omega$, then $\mathcal{F}_T = \mathcal{F}_{t_0}$.

Proof. Note that $\forall\ t \geq 0$,

$$\{T \leq t\} = \begin{cases} \varnothing, & t_0 > t \\ \Omega, & t_0 \leq t \end{cases}$$

"$\mathcal{F}_{t_0} \subseteq \mathcal{F}_T$"

$\forall\ A \in \mathcal{F}_{t_0}$, $\forall\ t \geq 0$,

① $t \geq t_0$, $A \cap \{T \leq t\} = A \in \mathcal{F}_{t_0} \subseteq \mathcal{F}_t$;

② $t < t_0$, $A \cap \{T \leq t\} = \varnothing \in \mathcal{F}_t$.

Hence $A \in \mathcal{F}_T \Rightarrow\ \mathcal{F}_{t_0} \subseteq \mathcal{F}_T$.

"$\mathcal{F}_T \subseteq \mathcal{F}_{t_0}$"

$\forall\ A \in \mathcal{F}_T$, we have $A \cap \{T \leq t\} \in \mathcal{F}_t$ by definition. Let $t = t_0$, then $\{T \leq t_0\} = \Omega$, so

$$A \cap \{T \leq t_0\} \in \mathcal{F}_{t_0} \Rightarrow\ A \cap \Omega \in \mathcal{F}_{t_0} \Rightarrow\ A \in \mathcal{F}_{t_0}.$$

Hence $A \in \mathcal{F}_{t_0} \Rightarrow\ \mathcal{F}_T \subseteq \mathcal{F}_{t_0}$ □

Lemma 1.3.3. $\forall$ stopping time T and S, $\forall\ A \in \mathcal{F}_S$, we have $A \cap \{S \leq T\} \in \mathcal{F}_T$. In particular, if $S \leq T$ for all $\omega \in \Omega$, then $\mathcal{F}_S \subseteq \mathcal{F}_T$.

Proof. $T \wedge t$ is also a stopping time. $\{T \wedge t \leq s\} \in \mathcal{F}_s \subseteq \mathcal{F}_t$. Hence $T \wedge t \in \mathcal{F}_t$.

To prove $A \cap \{S \leq T\} \in \mathcal{F}_T$, we need to prove $\forall\ t \geq 0$, $A \cap \{S \leq T\} \cap \{T \leq t\} \in \mathcal{F}_t$.

We have

$$A \cap \{S \leq T\} \cap \{T \leq t\} = \underbrace{A \cap \{S \leq t\}}_{\in \mathcal{F}_t} \cap \underbrace{\{T \leq t\}}_{\in \mathcal{F}_t} \cap \underbrace{\{S \wedge t \leq T \wedge t\}}_{\mathcal{F}_t}.$$

($S \leq T$ on $S \leq t$, $T \leq t \Leftrightarrow \{S \wedge t \leq T \wedge t\}$) □

Lemma 1.3.4. Let S, T be two stopping times. Then $\mathcal{F}_{T \wedge S} = \mathcal{F}_T \cap \mathcal{F}_S$ and $\{T < S\}$, $\{S < T\}$, $\{T \leq S\}$, $\{S \leq T\}$, $\{T = S\}$ belong to $\mathcal{F}_S \cap \mathcal{F}_T = \mathcal{F}_{S \wedge T}$.

Proof. $\mathcal{F}_{T \wedge S} \subseteq \mathcal{F}_S$ and $\mathcal{F}_{T \wedge S} \subseteq \mathcal{F}_T \Rightarrow\ \mathcal{F}_{T \wedge S} \subseteq \mathcal{F}_T \cap \mathcal{F}_S$.

To prove $\mathcal{F}_T \cap \mathcal{F}_S \subseteq \mathcal{F}_{T \wedge S}$, $\forall\ A \in \mathcal{F}_T \cap \mathcal{F}_S$, we will prove $\forall\ t \geq 0$, $A \cap \{T \wedge S \leq t\} \in \mathcal{F}_t$.

$$\begin{aligned} A \cap \{T \wedge S \leq t\} &= A \cap (\{T \leq t\} \cup \{S \leq t\}) \\ &= (A \cap \{T \leq t\}) \cup (A \cap \{S \leq t\}) \in \mathcal{F}_t. \end{aligned}$$

Next $\{S \leq T\} \in \mathcal{F}_T$, $\{S > t\} \in \mathcal{F}_T$. It remains to prove $\{S < T\} \in \mathcal{F}_T$.

Assuming this, by interchanging T and S, we get $\{T \leq S\}$, $\{T > S\}$, $\{T < S\} \in \mathcal{F}_S$.

Then $\{S \leq T\} \in \mathcal{F}_T$, $\{S > T\}^{\complement} \in \mathcal{F}_S$.

$\Rightarrow\ \{S \leq T\} \in \mathcal{F}_S \cap \mathcal{F}_T$.

$\{T = S\} = \{S \leq T\} \cap \{T \leq S\} \in \mathcal{F}_S \cap \mathcal{F}_T$.

To prove $\{S < T\} \in \mathcal{F}_T$, we define $R = S \wedge T$ such that $\{S < T\} = \{R < T\}$. Notice that $R \in \mathcal{F}_R \subseteq \mathcal{F}_T$. Hence $\{R < T\} \in \mathcal{F}_T$. □

Summary:

(i) $T \in \mathcal{F}_T$;

(ii) If $S \leq T$, $\mathcal{F}_S \subseteq \mathcal{F}_T$;

(iii) $T \wedge t \in \mathcal{F}_{T\wedge t} \subseteq \mathcal{F}_t$;

(iv) $\mathcal{F}_{T\wedge S} = \mathcal{F}_T \cap \mathcal{F}_S$.

For each $\omega \in \Omega$, $T(\omega) \in [0, \infty)$ and $X_t(\omega) \in \mathbb{R}^d$, $\forall\ t \geq 0$, we define $X_T(\omega) = X_{T(\omega)}(\omega)$.

Proposition 1.3.3. Let $\{X_t\}_{t\geq 0}$ be a progressively measurable process, Let T be a stopping time. Then X_T defined on $\{T < \infty\}$ is $\mathcal{F}_T$-measurable and the stopped process $\{X_{T\wedge t}\}_{t\geq 0}$ is progressively measurable.

Proof. $\forall\ B \in \mathcal{B}(\mathbb{R}^d)$, $\{X_T \in B\} \in \mathcal{F}_T \Leftrightarrow \forall\ t \geq 0$, $\underbrace{\{X_T \in B\} \cap \{T \leq t\}}_{=\{X_{T\wedge t} \in B\} \cap \{T \leq t\}} \in \mathcal{F}_t$.

We only need to prove $X_{T\wedge t} \in \mathcal{F}_t$.
Observe $(s, \omega) \to (T(\omega) \wedge s, \omega)$ is $\mathcal{B}([0, t]) \otimes \mathcal{F}_t$-measurable.
By assumption, $(s, \omega) \mapsto X_s(\omega)$ is $\mathcal{B}([0, t]) \otimes \mathcal{F}_t$-measurable.
So $(s, \omega) \to (T(\omega) \wedge s, \omega) \to (X_{T(\omega)\wedge s}, \omega)$ is $\mathcal{B}([0, t]) \otimes \mathcal{F}_t$-measurable. □

1.4 Cts-time martingale

Definition 1.4.1. For $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$, a martingale is an adapted stochastic process $(X_t)_{t\geq 0}$ such that

$$\begin{cases} (i)\ \forall\ t \geq 0,\ \mathbb{E}|X_t| < \infty \\ (ii)\ \forall\ t \geq s \geq 0,\ \mathbb{E}(X_t \mid \mathcal{F}_s) = X_s. \end{cases}$$

If $\mathbb{E}(X_t \mid \mathcal{F}_s) \geq X_s$, we call it a submartingale.
If $\mathbb{E}(X_t \mid \mathcal{F}_s) \leq X_s$, we call it a supmartingale.

Proposition 1.4.1. Let $\varphi : \mathbb{R} \to \mathbb{R}$ be a convex (concave up) function. If $(X_t)_{t\geq 0}$ is a martingale, then $(\varphi(X_t))_{t\geq 0}$ is a submartingale.

E.g. $|X_t|$, X_t^2, $|X_t|^p$ for all $p \geq 1$, $(X_t)^+$ are all submartingale.
We assume $\mathbb{E}\,|\varphi(X_t)| < \infty$, $\forall\ t \geq 0$.

Proof. Jensen's inequality. □

Theorem 1.4.1. Continuous-time Doob's upcrossing inequality

$$
\begin{aligned}
T_b^{(0)} &= -\infty, \\
T_a^{(1)} &= \inf\{t \geq (T_b^{(0)})^+ \colon X_t \leq a\}, \\
T_b^{(1)} &= \inf\{t \geq T_a^{(1)} \colon X_t \geq b\}.
\end{aligned}
$$

$\forall\ l \geq 2$,

$$
\begin{aligned}
T_a^{(l)} &= \inf\{t \geq T_b^{(l-1)} \colon X_t \leq a\}, \\
T_b^{l} &= \inf\{t \geq T_a^{l} \colon X_t \geq b\}
\end{aligned}
$$

Define $U_{ab}^X[0,n] = \max\{k \geq 0 \colon T_b^{(k)} \leq n\}$ to be the # of upcrossing in an interval [0,n].

Theorem 1.4.2. Let $(X_t)_{t\geq 0}$ be a right-continuous submartingale. For all $T > 0$,

$$
\mathbb{E}\left(U_{ab}^X[0,T]\right) \leq \frac{\mathbb{E}(X_T - a)^+}{b-a}
$$

Proof. For each $n \geq 1$, we let $D_n = \mathbb{Z}/2^n = \{0, \frac{\pm 1}{2^n}, \frac{\pm 2}{2^n}, \cdots\}$.

Consider the upcrossing of X on $[0,T] \cap D_n$, $\left\{\frac{X_j}{2^n}\right\}_{j=0}^{\infty}$ is a discrete-time martingale. Then $U_{ab}^X\left([0,T] \cap D_n\right) \leq U_{ab}^X[0,T]$, $U_{ab}^X\left([0,T] \cap D_n\right) \uparrow U_{ab}^X([0,T])$.

By using discrete-time upcrossing

$$
\mathbb{E}\left(U_{ab}^X\left([0,T] \cap D_n\right)\right) \leq \frac{1}{b-a}\mathbb{E}(X_{T_n} - a)^+,
$$

where $T_n = \frac{[T \cdot 2^n]}{2^n}$. Let $n \to \infty$ to get

$$
\mathbb{E}\left(U_{ab}^X([0,T])\right) \leq \frac{1}{b-a}\mathbb{E}(X_T - a)^+.
$$

□

Proposition 1.4.2. If $(X_t)_{t\geq 0}$ is a right continuous sub-martingale, and $\sup_{t\geq 0} \mathbb{E}X_t^+ < \infty$, then $\exists\ X \in L^1$ s.t. $X_t \xrightarrow{a.s.} X$ as $t \to \infty$.

Proof.

$$
\begin{aligned}
\mathbb{E}\left(U_{ab}^X([0,\infty))\right) &= \lim_{n\to\infty} \mathbb{E}\left(U_{ab}^X([0,n])\right) \\
&\leq \limsup_{n\to\infty} \frac{1}{b-a}\mathbb{E}(X_n - a)^+ \\
&\leq \limsup_{n\to\infty} \frac{1}{b-a}\left(\mathbb{E}(X_n^+) + |a|\right) < \infty.
\end{aligned}
$$

Hence $\forall\ a < b$, $U_{ab}^X[0,\infty) < \infty$ a.s.

$\Rightarrow$ w.p.1, $\forall\ a < b \in \mathbb{Q}$, $U_{ab}^X[0,\infty) < \infty$.

Hence $\liminf_{t\to\infty} X_t \leq a \leq b < \limsup_{t\to\infty} X_t$ does not occur.

$\Leftrightarrow \liminf_{t\to\infty} X_t = \limsup_{t\to\infty} X_t$. □

Definition 1.4.2. A family of random variables $(X_t)_{t\geq 0}$ is uniformly integrable (U.I.) if

$$\lim_{M\to\infty} \sup_{t\geq 0} \mathbb{E}\left(|X_t| \, 1_{|X_t|>M}\right) = 0.$$

Remark. If $\sup_{t\geq 0} \mathbb{E}\,|X_t|^p < \infty$ for some $p > 1$, then U.I.

Theorem 1.4.3. If $X_n \xrightarrow{P} X$, then equivalent

$$\begin{cases} X_n \xrightarrow{L^1} X, \\ \mathbb{E}\,|X_n| \to \mathbb{E}\,|X|\,, \\ \{X_n\} \text{ is U.I.}. \end{cases}$$

Proposition 1.4.3. Let $Z \in L^1$. Then $\{\mathbb{E}(Z \mid \mathcal{G})\colon \mathcal{G} \text{ is a sub } \sigma\text{-field of } \mathcal{F}\}$ is U.I.

In particular $\{\mathbb{E}(Z \mid \mathcal{F}_t)\}_{t\geq 0}$ is U.I. and it is a martingale. (By Tower's property)

Proposition 1.4.4. A right continuous martingale $(X_t)_{t\geq 0}$ is U.I. if and only if $\exists\, X_\infty \in L^1$ such that $X_t = \mathbb{E}(X_\infty \mid \mathcal{F}_t)$.

Proof. "$\Rightarrow$" U.I. implies $\sup_{t\geq 0} \mathbb{E}\,|X_t| < \infty$.

Then the martingale convergence theorem implies that $\exists\, X_\infty$ s.t. $X_n \xrightarrow{a.s.} X$ and $\mathbb{E}\,|X_\infty| = \mathbb{E}\left(\lim_{t\to\infty} |X_t|\right) \leq \liminf_{t\to\infty} \mathbb{E}\,|X_t| < \infty$.

To prove $\forall\, t \geq 0$, $X_t = \mathbb{E}(X_\infty \mid \mathcal{F}_t) \Leftrightarrow \forall\, A \in \mathcal{F}_t$, $\mathbb{E}(X_t 1_A) = \mathbb{E}(X_\infty 1_A)$.

For any $s \geq 0$, we have $\mathbb{E}(X_{t+s} \mid \mathcal{F}_t) = X_t \implies \mathbb{E}(X_{t+s} 1_A) = \mathbb{E}(X_t 1_A)$

$$|\mathbb{E}(X_{t+s} 1_A) - \mathbb{E}(X_\infty 1_A)| \leq \mathbb{E}\,|X_{t+s} - X_\infty|\,|1_A| \leq \mathbb{E}\,|X_{t+s} - X_\infty| \to 0.$$

Let $s \to \infty$, $\mathbb{E}(X_{t+s} 1_A) \xrightarrow{a.s.} \mathbb{E}(X_\infty 1_A)$.

Recall U.I. + a.s. $\implies L^1$ convergence.

"$\Leftarrow$" If $\exists\, X_\infty \in L^1$ s.t. $X_t = \mathbb{E}(X_\infty \mid \mathcal{F}_t)$, then $\{X_t\}_{t\geq 0}$ is U.I. $\square$

Theorem 1.4.4 (Optional sampling Theorem). Let $(X_t)_{t\geq 0}$ be a right continuous martingale. Let $S \leq T$ be two stopping time. Suppose either

① $\{X_t\}$ U.I.,

② $\exists$ constant $K > 0$, $S, T \leq K$.

Then $\mathbb{E}(X_T \mid \mathcal{F}_S) = X_S \implies \mathbb{E}(X_T) = \mathbb{E}(X_S) = \mathbb{E}(X_0)$.

Proof. Define ① $Z = X_\infty$, ② $Z = X_K$.

In either case, $X_t = \mathbb{E}(Z \mid \mathcal{F}_t)$, ① $\forall\, t \geq 0$, ② $\forall\, t \leq K$.

It suffices to prove $X_T = \mathbb{E}(Z \mid \mathcal{F}_T)$. Then $X_S = \mathbb{E}(Z \mid \mathcal{F}_S)$.

Hence

$$\begin{aligned}\mathbb{E}(X_T \mid \mathcal{F}_S) &= \mathbb{E}\left(\mathbb{E}(Z \mid \mathcal{F}_T) \mid \mathcal{F}_S\right) \\ &= \mathbb{E}(Z \mid \mathcal{F}_S) \\ &= X_S.\end{aligned}$$

① Suppose $T \in \{t_1, t_2, \dots\}$. Then $\forall\ A \in \mathcal{F}_T$,

$$\begin{aligned}\mathbb{E}(X_T \mathbf{1}_A) &= \mathbb{E}(Z \mathbf{1}_A) \\ &= \sum_{n=1}^{\infty} \mathbb{E}(X_{t_n} \mathbf{1}_{\{T=t_n\}} \mathbf{1}_A) \\ &= \sum_{n=1}^{\infty} \mathbb{E}\left(\mathbf{1}_{A \cap \{T=t_n\}} Z\right) \\ &= \mathbb{E}(\mathbf{1}_A Z)\end{aligned}$$

because ① $A \cap \{T = t_n\} \in \mathcal{F}_{t_n}$, ② $\mathbb{E}(Z \mid \mathcal{F}_{t_n}) = X_{t_n}$

② General T. Define $T_k = \frac{[2^k T]+1}{2^k}$.

To prove $X_T = \mathbb{E}(Z \mid \mathcal{F}_T)$, $\forall\ A \in \mathcal{F}_T \subseteq \mathcal{F}_{T_k}$, we get $\mathbb{E}(Z \mid \mathcal{F}_{T_k}) = X_{T_k} \implies \mathbb{E}(Z\mathbf{1}_A) = \mathbb{E}(X_{T_k}\mathbf{1}_A)$.

Send $k \to \infty$ to see $\mathbb{E}(Z\mathbf{1}_A) = \mathbb{E}(X_T\mathbf{1}_A)$.

Because ① $X_{T_k} \xrightarrow{a.s.} X_T$ by right-continuity, ② $\{X_{T_k}\}$ is U.I. by $X_{T_k} = \mathbb{E}(Z \mid \mathcal{F}_{T_k})$, ③ $\implies X_{T_k} \xrightarrow{L^1} X_T$. $\square$

Definition 1.4.3 (defn example)**.** Example

Theorem 1.4.5 (thm example)**.** Example

Proof. proof □

Lemma 1.4.1 (lemma example)**.** Example

Proof for Lemma.

■. proof for lemma example □

Proposition 1.4.5. pro example

Proof for Proposition.

■. Example □

Claim 1.4.1. claim example Example

Proof for Claim.

■. proof for claim example □

Corollary 1.4.1 (cor example)**.** example

Proof for Corollary.

■. proof for corollary example □

Fact 1.4.1. Example fact

Remark. Example remark

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